

Reduced Basis methods: an introduction

Christophe Prud'homme
prudhomme@unistra.fr

September 12, 2023

CeMosis - <http://www.cemosis.fr>
IRMA
Université de Strasbourg

① SVD and POD

SVD and POD

Singular Value Decomposition

Proposition

For each matrix $Y \in \mathbb{R}^{m \times n}$ there exists orthogonal matrices $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$ and a diagonal matrix

$$\Sigma := \text{diag}(\sigma_1, \dots, \sigma_p) \in \mathbb{R}^{m \times n}, \quad p = \min(m, n),$$

where

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_p \geq 0$$

such that

$$Y = U \Sigma V^T$$

Here:

- Singular values of Y : $\sigma_i, i = 1, \dots, p$
- Left singular vectors : columns of $U = [u_1 u_2 \dots u_m]$
- Right singular vectors: columns of $V = [v_1 v_2 \dots v_n]$

SVD

Case: $m = 4, n = 2, \text{rank}(Y) = 2$

- Full SVD

$$\begin{bmatrix} \times & \times \\ \times & \times \\ \times & \times \\ \times & \times \end{bmatrix} = \begin{bmatrix} \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \end{bmatrix} \begin{bmatrix} \times & 0 \\ 0 & \times \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \times & \times \\ \times & \times \end{bmatrix}$$

(1)

- Reduced SVD

$$\begin{bmatrix} \times & \times \\ \times & \times \\ \times & \times \\ \times & \times \end{bmatrix} = \begin{bmatrix} \times & \times \\ \times & \times \\ \times & \times \\ \times & \times \end{bmatrix} \begin{bmatrix} \times & 0 \\ 0 & \times \end{bmatrix} \begin{bmatrix} \times & \times \\ \times & \times \end{bmatrix} \quad (2)$$

SVD

- Assumption: $m \geq n$
- Case $m < n$: consider SVD of Y^T

$$Y^T = U \Sigma V^T \Rightarrow Y = V U^T$$

- Singular values are real and non-negative
- Convention
 - $\sigma_{\max} = \sigma_1$ largest singular value
 - $\sigma_{\min} = \sigma_n$ smallest singular values
 - ordered according to magnitude

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n \geq 0$$

- Numerical computation of SVD is backward stable

$$Y + \Delta Y = \tilde{U} \tilde{\Sigma} \tilde{V}^T \text{ with } \|\Delta Y\|_2 = \mathcal{O}(\text{eps})$$

SVD - Geometry

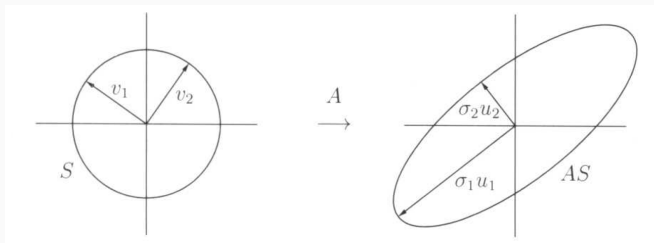
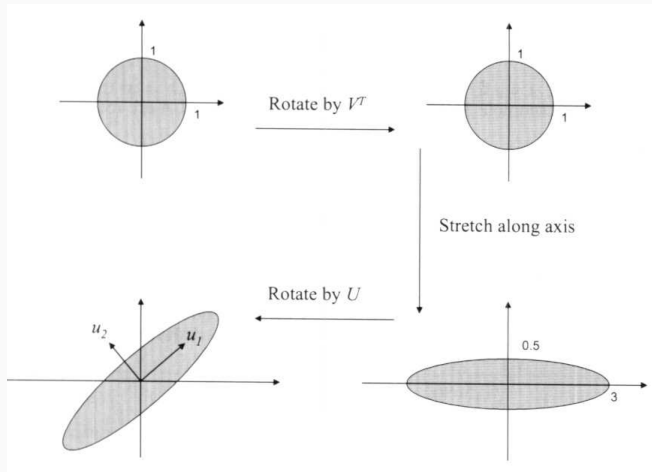


Figure 1: Mapping of unit sphere $S = \{x : \|x\|_2 = 1\}$ under Y

- Singular values: lengths σ_1, σ_2 of principal semi-axes of YS .
- Left singular vectors: unit vectors $\{u_1, u_2\}$ in direction of principal semi-axes of YS
- Right singular vectors: unit vectors $\{v_1, v_2\} \in S$, preimages of principal semi-axes of YS , so that $Yv_j = \sigma_j u_j$.

SVD - Geometry



Properties

Let $\mathbf{Y} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ be an SVD of $\mathbf{Y} \in \mathbb{R}^{m \times n}$ with singular values $\sigma_1 \geq \dots \geq \sigma_d \geq \sigma_{d+1} = \dots = \sigma_n = 0$. We then have:

- $\mathbf{Y}\mathbf{v}_i = \sigma_i\mathbf{u}_i, \mathbf{Y}^T\mathbf{u}_i = \sigma_i\mathbf{v}_i, i = 1, \dots, n$.
- $\text{Rang}(\mathbf{Y}) = \mathbf{d}$ = number of singular values not equal to zero.
- $\|\mathbf{Y}\|_2 = \sigma_1 = \sigma_{\max}$
- $\|\mathbf{Y}\|_F = \sqrt{\sigma_1^2 + \dots + \sigma_n^2}$
- $\|\mathbf{Y}^{-1}\|_2 = \frac{1}{\sigma_n}$, if $\mathbf{Y} \in \mathbb{R}^{n \times n}$ regular
- $\kappa_2(\mathbf{Y}) = \|\mathbf{Y}\|_2 \|\mathbf{Y}^{-1}\|_2 = \frac{\sigma_1}{\sigma_n}$, if $\mathbf{Y} \in \mathbb{R}^{n \times n}$ regular
- $\kappa_2(\mathbf{Y}) = \|\mathbf{Y}\|_2 \|\mathbf{Y}^+\|_2 = \frac{\sigma_1}{\sigma_n} = \frac{\sigma_{\max}}{\sigma_{\min}}$, if $\mathbf{Y}$ rectangular
- $\sigma_i^2 = \lambda_i$, where λ_i are the eigenvalues of $\mathbf{Y}^T\mathbf{Y}$ (a square matrix).
- Reduced SVD: in case $m > n$ (frequent in data science), then $\mathbf{\Sigma}$ has at most n non-zero singular values. The reduced SVD (or economy-sized) is then given by

$$\mathbf{Y} = \hat{\mathbf{U}}\hat{\mathbf{\Sigma}}\mathbf{V}^T$$

where $\hat{\mathbf{\Sigma}}$ is a $n \times n$ diagonal matrix

Computation of an SVD: Octave/Matlab, Python

All SVD implementations are based on the LAPACK routine dgesvd with a few exceptions

```
A=rand(5,3) % random 5x3 matrix
[U,S,V] = svd(A) % full SVD
[Uh,Sh,Vh] = svd(A,'econ') % reduced SVD
```

```
import numpy as np
A=np.random.rand(5,3) # random 5x3 matrix
U,S,V = np.linalg.svd(A) # full SVD
# full SVD
U,S,V = np.linalg.svd(A,full_matrices=True)
# reduced SVD
Uh,Sh,Vh = np.linalg.svd(A,full_matrices=False)
```

Properties

Let $Y = U\Sigma V^T$ be an SVD of $Y \in \mathbb{R}^{m \times n}$ with singular values $\sigma_1 \geq \dots \geq \sigma_d \geq \sigma_{d+1} = \dots = \sigma_n = 0$. We then have:

- The strictly positive singular values are the roots of the strictly positive eigenvalues of $Y^T Y$:

$$\{\sigma_i \mid i = 1, \dots, d\} = \left\{ \sqrt{\lambda_i(Y^T Y)} \mid i = 1, \dots, d \right\}$$

- The singular values are equal to the absolute values of the eigenvalues of Y if $Y = Y^T$.
- For $Y \in \mathbb{R}^{n \times n}$ it holds $|\det(Y)| = \prod_{i=1}^n \sigma_i$
- The pseudo inverse $Y^+ \in \mathbb{R}^{n \times m}$ is defined as $Y^+ = V\Sigma^+ U^T$ where $\Sigma^+ = \text{diag}(\sigma_1^{-1}, \dots, \sigma_d^{-1}, 0, \dots, 0) \in \mathbb{R}^{n \times m}$.

Properties - Low Rank Approximation

Let $\mathbf{Y} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ be an SVD of $\mathbf{Y} \in \mathbb{R}^{m \times n}$ with singular values $\sigma_1 \geq \dots \geq \sigma_d \geq \sigma_{d+1} = \dots = \sigma_n = 0$. For $k \leq d = \text{Rang}(\mathbf{Y})$ define

$$\mathbf{Y}_k = \mathbf{U}\mathbf{\Sigma}_k\mathbf{V}^T$$

where $\mathbf{\Sigma}_k = \text{diag}(\sigma_1, \dots, \sigma_k, 0, \dots, 0) \in \mathbb{R}^{m \times n}$. We then have:

- $\text{Rang}(\mathbf{Y}_k) = k$.
- The distance between $\mathbf{Y}_k$ and $\mathbf{Y}$ in the 2-norm is

$$\|\mathbf{Y} - \mathbf{Y}_k\|_2 = \sigma_{k+1}.$$

- $\mathbf{Y}_k$ is the best approximation of $\mathbf{Y}$ of rank $\leq k$

$$\|\mathbf{Y} - \mathbf{Y}_k\|_2 = \min_{\text{Rang}(B) \leq k} \|\mathbf{Y} - B\|_2$$

Application : Low Rank Approximation

Singular values decomposition of $\mathbf{Y}$

$$\mathbf{Y} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

can be written as

$$\mathbf{Y} = \sum_{i=1}^p \sigma_i \mathbf{u}_i \mathbf{v}_i^T = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \dots + \sigma_p \mathbf{u}_p \mathbf{v}_p^T$$

where $p = \min(\mathbf{m}, \mathbf{n})$.

Approximation of matrix $\mathbf{Y}$ of rank $\mathbf{k}$, $\mathbf{k} \leq \mathbf{p}$, is

$$\mathbf{Y}_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^T$$

and $\|\mathbf{Y} - \mathbf{Y}_k\|_2 = \sigma_{k+1}$

Data compression:

$\Rightarrow$ Memory requirement of $\mathbf{Y}_k$ is $k(m + n)$ vs. mn for $\mathbf{Y}$.

How many singular values are needed?

Numerous factor could intervene depending on the problem

- the desired rank of the system
- the noise amplitude
- the distribution of the singular values

A very common approach is to choose the value of the rank r that captures a given variance or energy of the data. In [Donoho¹] we have a theoretical result that provides a hard threshold, τ_* .

¹M. GAVISH AND D. L. DONOHO, The optimal hard threshold for singular values is $4/\sqrt{3}$, IEEE Transactions on Information Theory, 60 (2014), pp. 5040-5053, <https://doi.org/10.1109/TIT.2014.2323359>.

Hard threshold decision tree I

Suppose X is a matrix of interest that is suspected to be low-rank and that we have a noisy observation $Y = X + \gamma Z$ where the noise matrix Z has i.i.d. entries with zero mean and unit variance, and γ is the noise amplitude.

- if γ is not known, then use the median of the singular values as a threshold $\tau^* = \omega(\beta) * \hat{\sigma}$
- if γ is known then if X is not square then use $\tau_* = \lambda(\beta)\sqrt{n}\gamma$
- if γ is known then if X is square then use $\tau_* = \frac{4}{\sqrt{3}}\sqrt{n}\gamma$

with In the nonsquare case, for $m \ll n$ we define the aspect ratio $\beta = \frac{m}{n}$,

$$\text{and } \lambda(\beta) = \left[2(\beta + 1) + \frac{8\beta}{(\beta+1)+(\beta^2+14\beta+1)^{1/2}} \right]^{1/2}.$$

Hard threshold decision tree II

Note that this reduces to $4/\sqrt{3}$ when $\beta = 1$ and when $n \ll m$ we must define $\beta = n/m$ so that $0 < \beta < 1$ always.

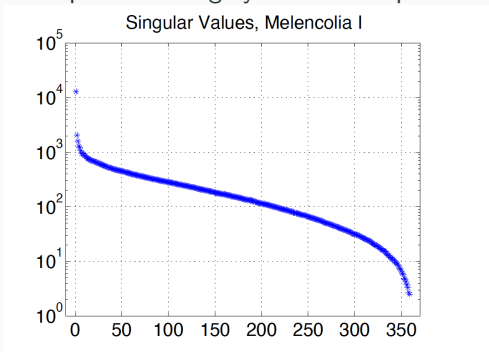
When γ is unknown, we estimate it from the data by the median of the singular values, $\hat{\sigma}$, and define $\omega(\beta) \approx 0.56\beta^3 - 0.95\beta^2 + 1.82\beta + 1.43$ for rectangular matrices, which gives $\tau_* \approx 2.858\hat{\sigma}$ for a square matrix ($\beta = 1$).

Optimal SVD Thresholding

See Notebook on Singular Value Thresholding.

Image Compression

Consider black and white picture $I(x, y)$ as a matrix Y : matrix entries correspond to the gray-level of the pixels.

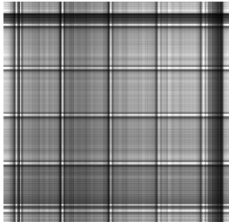


Note

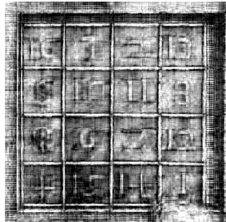
Depending on the area of application: Principal Component Analysis (PCA), Proper Orthogonal Decomposition (POD), Karhunen-Loève Decomposition.

Image Compression

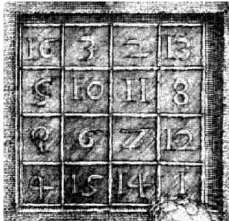
$k = 1$, compression = 0.00548



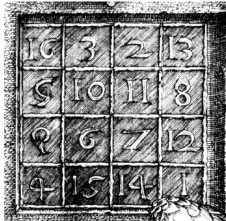
$k = 20$, compression = 0.11



$k = 40$, compression = 0.219



Original: A. Durer, Melencolia I



SVD: Further reading

Recently in [REF²] the previous results have been extended to the case of correlated noise.

²D. L. DONOHO, M. GAVISH, AND E. ROMANOV, *Code supplement for "ScreenOT: Exact MSE-optimal singular value thresholding in correlated noise"*, report, Stanford Digital Repository, 2020, <https://arxiv.org/abs/2009.12297>, <https://purl.stanford.edu/py>

SVD and POD I

We express the reduced SVD of Y as

$$Y = U^d D (V^d)^T,$$

where D is the reduced diagonal matrix $d \times d$, $U^d \in \mathbb{R}^{m \times d}$ and $V^d \in \mathbb{R}^{n \times d}$ are given by

$$\begin{aligned} U_{ij}^d &= U_{ij} & \text{for } 1 \leq i \leq m, 1 \leq j \leq d \\ V_{ij}^d &= V_{ij} & \text{for } 1 \leq i \leq n, 1 \leq j \leq d \end{aligned}$$

Setting $B^d = D (V^d)^T \in \mathbb{R}^{d \times n}$ we can rewrite the SVD in the form

$$Y = U^d B^d \quad \text{with } B^d = D (V^d)^T \in \mathbb{R}^{d \times n}.$$

Thus, the column space of Y can be represented in terms of the d

SVD and POD II

linearly independent columns of U^d . The coefficients in the expansion for the columns y_j , $j = 1, \dots, n$, in the basis $\{u_i\}_{i=1}^d$ are given by the j th-column of B^d . Since U is orthogonal, we find that

$$\begin{aligned}
 y_j &= \sum_{i=1}^d B_{ij}^d U_{:,i}^d = \sum_{i=1}^d \left(D (V^d)^T \right)_{ij} u_i = \sum_{i=1}^d \underbrace{\left((U^d)^T U^d D (V^d)^T \right)_{ij}}_{=I^d \in \mathbb{R}^{d \times d}} u_i \\
 &\stackrel{\text{SVD}}{=} \sum_{i=1}^d \left((U^d)^T Y \right)_{ij} u_i = \sum_{i=1}^d \underbrace{\left(\sum_{k=1}^m U_{ki}^d Y_{kj} \right)}_{=u_i^T y_j} u_i = \sum_{i=1}^d \langle u_i, y_j \rangle_{\mathbb{R}^m} u_i,
 \end{aligned}$$

where $\langle \cdot, \cdot \rangle_{\mathbb{R}^m}$ denotes the canonical inner product in $\mathbb{R}^m$. Thus,

$$y_j = \sum_{i=1}^d \langle y_j, u_i \rangle_{\mathbb{R}^m} u_i \quad \text{for } j = 1, \dots, n$$

SVD and POD

We consider the problem of approximating all spatial vectors $\{\mathbf{y}_j\}_{j=1}^n$ of $\mathbf{Y}$ simultaneously by a single, normalized vector as well as possible.

This problem can be expressed as

$$(P^1) \quad \max_{u \in \mathbb{R}^m} \sum_{j=1}^n |(\mathbf{y}_j, u)_{\mathbb{R}^m}|^2$$

subject to (s.t.) $\|u\|_{\mathbb{R}^m}^2 = 1$

Solution: u_1 solves (P^1) and $\arg \max (P^1) = \sigma_1^2 = \lambda_1$.

Find second vector, orthogonal to u_1 , that describes $\{\mathbf{y}_j\}_{j=1}^n$ as well as possible

$$(P^2) \quad \max_{u \in \mathbb{R}^m} \sum_{j=1}^n |(\mathbf{y}_j, u)_{\mathbb{R}^m}|^2$$

s.t. $\|u\|_{\mathbb{R}^m}^2 = 1, (u, u_1)_{\mathbb{R}^m} = 0$

Solution: u_2 solves (P^2) and $\arg \max (P^2) = \sigma_2^2 = \lambda_2$.

SVD and POD

Theorem

Let $Y = [y_1, \dots, y_n] \in \mathbb{R}^{m \times n}$ be a given matrix with rank $d \leq \min\{m, n\}$. Further, let $Y = U\Sigma V^T$ be the singular value decomposition of Y , where $U = [u_1, \dots, u_m] \in \mathbb{R}^{m \times m}$, $V = [v_1, \dots, v_n] \in \mathbb{R}^{n \times n}$ are orthogonal matrices and the matrix $\Sigma \in \mathbb{R}^{m \times n}$ contains the singular values. Then, for any $\ell \in \{1, \dots, d\}$ the solution to

$$\begin{aligned}
 (P^\ell) \quad & \max_{\tilde{u}_1, \dots, \tilde{u}_\ell \in \mathbb{R}^m} \sum_{i=1}^{\ell} \sum_{j=1}^n |(y_j, \tilde{u}_i)_{\mathbb{R}^m}|^2 \\
 \text{s.t.} \quad & (\tilde{u}_i, \tilde{u}_j)_{\mathbb{R}^m} = \delta_{ij}, 1 \leq i, j \leq \ell
 \end{aligned}$$

is given by the singular vector $\{u_i\}_{i=1}^{\ell}$, i.e., by the first ℓ columns of U . Moreover,

$$\arg \max (P^\ell) = \sum_{i=1}^{\ell} \sigma_i^2 = \sum_{i=1}^{\ell} \lambda_i$$

Optimality of the POD basis

Theorem

Optimality³ Suppose that $\hat{U}^d \in \mathbb{R}^{m \times d}$ denotes a matrix with pairwise orthonormal vectors $\hat{u}_i$ and that the expansion of the columns of Y in the basis $\{\hat{u}_i\}_{i=1}^d$ be given by $Y = \hat{U}^d C^d$, with $C_{ij}^d = (\hat{u}_i, y_j)_{\mathbb{R}^m}$, $1 \leq i \leq d, 1 \leq j \leq n$. Then for every $\ell \in \{1, \dots, d\}$ we have

$$\|Y - U^\ell B^\ell\|_F \leq \|Y - \hat{U}^\ell C^\ell\|_F$$

where $\|\cdot\|_F$ denotes the Frobenius norm, the matrix U^ℓ denotes the first ℓ columns of U , B^ℓ the first ℓ rows of B and similarly for $\hat{U}^\ell$ and C^ℓ

³S. Volkwein. Model Reduction Using Proper Orthogonal Decomposition. Skript Universität Konstanz, 2010.

POD principle

POD, or *Proper Orthogonal Decomposition*, is a method for reducing the dimensionality of a dataset by representing it using an orthonormal basis and optimal in a least-squares sense. The original variables are replaced with a new set of uncorrelated variables, called POD modes or principal components, with the first few modes typically retaining most of the data's energy. By keeping only the first few modes, a lower dimensional representation of the data is obtained.

Other Names

In the field of stochastic processes, this method is known as **Karhunen-Loève (KL) decomposition**, and in multivariate statistics it is known as **Principal Component Analysis**.

POD

POD is widely used in the field of reduced-order modeling for PDEs.

A set of n_s parameter samples, $\Xi_s = \{\boldsymbol{\mu}^1, \dots, \boldsymbol{\mu}^{n_s}\}$, and corresponding solutions, $\{u_h(\boldsymbol{\mu}^1), \dots, u_h(\boldsymbol{\mu}^{n_s})\}$, of the high-fidelity problem are used to define the snapshot matrix, $\mathbb{Z} \in \mathbb{R}^{\mathcal{N} \times n_s}$, as: $\mathbb{Z} = [u_1 | \dots | u_{n_s}]$.

The vectors $u_i \in \mathbb{R}^{\mathcal{N}}$, $1 \leq i \leq n_s$, represent the degrees of freedom of the functions $u^{\mathcal{N}}(\boldsymbol{\mu}^i) \in V_h$, where $u_i^{(j)} = u_h^{(j)}(\boldsymbol{\mu}^i)$ for $1 \leq i \leq n_s$ and $1 \leq j \leq \mathcal{N}$.

POD using the ℓ_2 inner product

The SVD decomposition of $\mathbb{Z}$ reads

$$\mathbb{Z} = \mathbb{U}\Sigma\mathbb{S}^T,$$

where $\mathbb{U} = [\zeta_1 | \dots | \zeta_{\mathcal{N}}] \in \mathbb{R}^{\mathcal{N} \times \mathcal{N}}$ and $\mathbb{S} = [\psi_1 | \dots | \psi_{n_s}] \in \mathbb{R}^{n_s \times n_s}$ are orthogonal matrices, and $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r) \in \mathbb{R}^{\mathcal{N} \times n_s}$ with $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r$. Here $r \leq \min(\mathcal{N}, n_s)$ denotes the rank of $\mathbb{Z}$, which is strictly smaller than n_s if the snapshot vectors are not all linearly independent. Then, we can write

$$\mathbb{Z}\psi_i = \sigma_i\zeta_i \quad \text{and} \quad \mathbb{Z}^T\zeta_i = \sigma_i\psi_i, \quad i = 1, \dots, r \quad (3)$$

or, equivalently,

$$\mathbb{Z}^T\mathbb{Z}\psi_i = \sigma_i^2\psi_i \quad \text{and} \quad \mathbb{Z}\mathbb{Z}^T\zeta_i = \sigma_i^2\zeta_i, \quad i = 1, \dots, r \quad (4)$$

Correlation matrix

The matrix $\mathbb{C} = \mathbb{Z}^T \mathbb{Z} \in \mathbb{R}^{n_s \times n_s}$ is called **correlation matrix** given by:

$$\mathbb{C}_{ij} = \mathbf{u}_i^T \mathbf{u}_j, \quad 1 \leq i, j \leq n_s.$$

For any $N \leq n_s$, the POD basis $\mathbb{V} \in \mathbb{R}^{\mathcal{N} \times N}$ of dimension N is defined as the set of the first N left singular vectors $\zeta_1, \dots, \zeta_N$ of $\mathbb{U}$ or using (3)

$$\zeta_j = \frac{1}{\sigma_j} \mathbb{Z} \psi_j, \quad 1 \leq j \leq N$$

the set of vectors obtained from the first N eigenvectors $\psi_1, \dots, \psi_N$ of $\mathbb{C}$. Two properties:

- By construction, the POD basis is orthonormal
- The POD basis minimizes, over all possible N -dimensional orthonormal bases $\mathbb{W} = [\mathbf{w}_1 | \dots | \mathbf{w}_N] \in \mathbb{R}^{\mathcal{N} \times N}$, the sum of the squares of the errors between each snapshot vector $\mathbf{u}_i$ and its projection onto the subspace spanned by $\mathbb{W}$

Proposition I

Recall that the projection $\Pi_W x$ of a vector $x \in \mathbb{R}^N$ onto $\text{span}(W)$ is given by

$$\Pi_W x = \sum_{j=1}^N (x, w_j)_2 w_j = W^T x$$

Proposition

Let $\mathcal{V}_N = \{W \in \mathbb{R}^{N \times N} : W^T W = \mathbb{I}_N\}$ be the set of all N dimensional orthonormal bases. Then,

$$\sum_{i=1}^{n_s} \|u_i - VV^T u_i\|_2^2 = \min_{W \in \mathcal{V}_N} \sum_{i=1}^{n_s} \|u_i - WW^T u_i\|_2^2 = \sum_{i=N+1}^r \sigma_i^2$$

Proposition II

Proof. The best rank N approximation of $\mathbb{Z}$ is given by

$$\mathbb{Z}_N = \sum_{i=1}^N \sigma_i \zeta_i \psi_i^T.$$

From the second relation in 3, we have that

$$\psi_i = \frac{1}{\sigma_i} \mathbb{Z}^T \zeta_i$$

and therefore

$$\mathbb{Z}_N = \sum_{i=1}^N \sigma_i \zeta_i \frac{1}{\sigma_i} (\mathbb{Z}^T \zeta_i)^T = \sum_{i=1}^N \zeta_i (\zeta_i^T \mathbb{Z}) = \mathbb{V} \mathbb{V}^T \mathbb{Z}$$

Proposition III

For a matrix $\mathbb{A} \in \mathbb{R}^{m \times n}$ we have: $\|\mathbb{A}\|_F^2 = \sum_{i=1}^n \|a_i\|_2^2$, where a_i denotes the i -th column of $\mathbb{A}$. Thus, for $\mathbb{W} \in \mathcal{V}_N$ we can write

$$\sum_{i=1}^{n_s} \left\| u_i - \mathbb{W} \mathbb{W}^T u_i \right\|_2^2 = \left\| \mathbb{Z} - \mathbb{W} \mathbb{W}^T \mathbb{Z} \right\|_F^2.$$

Since

$$\|\mathbb{Z} - \mathbb{Z}_N\|_F^2 = \min_{\mathbb{B} \in \mathbb{R}^{N \times n_s}, \text{rank}(\mathbb{B}) \leq N} \|\mathbb{Z} - \mathbb{B}\|_F^2 \leq \min_{\mathbb{W} \in \mathcal{V}_N} \|\mathbb{Z} - \mathbb{W} \mathbb{W}^T \mathbb{Z}\|_F^2$$

and $\mathbb{Z}_N = \mathbb{V} \mathbb{V}^T \mathbb{Z}$, we conclude that

$$\|\mathbb{Z} - \mathbb{V} \mathbb{V}^T \mathbb{Z}\|_F^2 = \min_{\mathbb{W} \in \mathcal{V}_N} \|\mathbb{Z} - \mathbb{W} \mathbb{W}^T \mathbb{Z}\|_F^2 \quad \square$$

Relative Information Content

The error in the POD basis is determined by summing the squares of the singular values associated with ignored POD modes. This outcome proposes an appropriate standard for selecting the minimal POD dimension $N \leq r$ such that the projection error is less than a required tolerance ε_{POD} . It is sufficient to choose N as the smallest integer that satisfies the condition.

$$I(N) = \frac{\sum_{i=1}^N \sigma_i^2}{\sum_{i=1}^r \sigma_i^2} \geq 1 - \varepsilon_{\text{POD}}^2,$$

that is the energy retained by the last $r - N$ modes is equal or smaller than $\varepsilon_{\text{POD}}^2$.

i $I(N)$ is referred to as the relative information content (RIC) of the POD basis. It represents the percentage of energy of the snapshots captured by the first N POD modes.

POD

Proper Orthogonal Decomposition (POD) Algorithm

- 1: **Input:** Snapshot matrix $\mathbb{Z} \in \mathbb{R}^{\mathcal{N} \times n_s}$
- 2: Compute the singular value decomposition of $\mathbb{Z}$: $\mathbb{Z} = \mathbb{U}\mathbf{\Sigma}\mathbb{S}^T$
- 3: Select the first N columns of $\mathbb{U}$ as the POD modes: $\mathbb{U}_N = \mathbb{U}(:, 1 : N)$
- 4: Project the snapshots onto the POD modes: $\mathbb{V} = \mathbb{U}_N^T \mathbb{Z}$
- 5: **Output:** POD modes $\mathbb{U}_N$ and coefficients $\mathbb{V}$

i Note

this is a basic version of the POD algorithm and there might be variations depending on the specific problem and context you're working with, and some extra steps might be added to the algorithm.

POD - alternative with the correlation matrix

Proper Orthogonal Decomposition (POD) Algorithm

- 1: Compute $\mathbb{V} = \text{POD}(\mathbb{Z}, \varepsilon_{\text{POD}})$
- 2: form the correlation matrix $\mathbb{C} = \mathbb{Z}^T \mathbb{Z}$
- 3: solve the eigenvalue problem $\mathbb{C}\psi_i = \sigma_i^2 \psi_i, \quad i = 1, \dots, r$
- 4: set $\zeta_i = \frac{1}{\sigma_i} \mathbb{Z} \psi_i$
- 5: define N as the minimum integer such that $I(N) \geq 1 - \varepsilon_{\text{POD}}$
- 6: set $\mathbb{V} = [\zeta_1 | \dots | \zeta_N]$

POD and the energy inner product

Since the snapshots functions $u_h(\mu^i)$ belong to $X^{\mathcal{N}} \subset X$, it is natural to seek an alternative POD basis which minimizes the X -norm ($\mathbb{X}$) - rather than the $\|\cdot\|_2$ norm - of the projection error of the snapshots vectors u_i .

We wish to seek a basis $\mathbb{W} \in \mathcal{V}_N^{X^{\mathcal{N}}}$, with

$$\mathcal{V}_N^{X^{\mathcal{N}}} = \{ \mathbb{W} \in \mathbb{R}^{N \times N} : \mathbb{W}^T \mathbb{X} \mathbb{W} = \mathbb{I}_N \},$$

which minimizes the squares of the $X^{\mathcal{N}}$ -norm of the error between each snapshot vector u_i and its $X^{\mathcal{N}}$ -orthogonal projection onto the subspace spanned by $\mathbb{W}$, i.e.

$$\min_{\mathbb{W} \in \mathcal{V}_N^{X^{\mathcal{N}}}} \sum_{i=1}^{n_s} \left\| u_i - \Pi_{\mathbb{W}}^{X^{\mathcal{N}}} u_i \right\|_X^2$$

with

$$\Pi_{\mathbb{W}}^{X^{\mathcal{N}}} x = \sum_{j=1}^N (x, w_j)_{\mathbb{X}} w_j = \mathbb{W} \mathbb{W}^T \mathbb{X} x$$

the $X^{\mathcal{N}}$ -orthogonal projection of $x \in \mathbb{R}^{\mathcal{N}}$ onto $\text{span}(\mathbb{W})$.

Recasting the problem

We have

$$\begin{aligned} \sum_{i=1}^{n_s} \|u_i - \Pi_W^{\mathbb{X}} u_i\|_{\mathbb{X}}^2 &= \sum_{i=1}^{n_s} \|u_i - \mathbb{W}^T \mathbb{W}^T \mathbb{X} u_i\|_{\mathbb{X}}^2 \\ &= \sum_{i=1}^{n_s} \left\| \mathbb{X}^{1/2} u_i - \mathbb{X}^{1/2} \mathbb{W}^T \mathbb{X} u_i \right\|_2^2 = \|\mathbb{X}^{1/2} \mathbb{Z} - \mathbb{X}^{1/2} \mathbb{W}^T \mathbb{X} \mathbb{Z}\|_F^2. \end{aligned}$$

Substituting and setting $\tilde{\mathbb{Z}} = \mathbb{X}^{1/2} \mathbb{Z}$ and $\tilde{\mathbb{W}} = \mathbb{X}^{1/2} \mathbb{W}$, we finally obtain

$$\min_{\mathbb{W} \in \mathcal{V}_N^{\mathbb{X}}} \sum_{i=1}^{n_s} \|u_i - \Pi_W^{\mathbb{X}_h} u_i\|_{\mathbb{X}_h}^2 = \min_{\tilde{\mathbb{W}} \in \mathcal{V}_N} \left\| \tilde{\mathbb{Z}} - \tilde{\mathbb{W}} \tilde{\mathbb{W}}^T \tilde{\mathbb{Z}} \right\|_F^2$$

Proposition

Let $\mathbb{Z} = [\mathbf{u}_1 | \dots | \mathbf{u}_{n_s}] \in \mathbb{R}^{\mathcal{N} \times n_s}$ be a given matrix of rank $r \leq \min(\mathcal{N}, n_s)$, $\mathbb{X} \in \mathbb{R}^{\mathcal{N} \times \mathcal{N}}$ a symmetric positive definite matrix, $\tilde{\mathbb{Z}} = \mathbb{X}^{1/2} \mathbb{Z}$ and $\tilde{\mathbb{Z}} = \tilde{\mathbb{U}} \Sigma \tilde{\mathbb{Z}}^T$ its singular value decomposition, where

$$\tilde{\mathbb{U}} = [\tilde{\zeta}_1 | \dots | \tilde{\zeta}_{\mathcal{N}}] \in \mathbb{R}^{\mathcal{N} \times \mathcal{N}}, \quad \tilde{\mathbb{Z}} = [\tilde{\psi}_1 | \dots | \tilde{\psi}_{n_s}] \in \mathbb{R}^{n_s \times n_s}$$

are orthogonal matrices and $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r) \in \mathbb{R}^{\mathcal{N} \times n_s}$ with $\sigma_1 \geq \dots \geq \sigma_r$. Then, for $N \leq r$, the POD basis $\mathbb{V} = [\mathbb{X}^{-1/2} \tilde{\zeta}_1 | \dots | \mathbb{X}^{-1/2} \tilde{\zeta}_N]$ is such that

$$\sum_{i=1}^{n_s} \left\| \mathbf{u}_i - \mathbb{V} \mathbb{V}^T \mathbb{X} \mathbf{u}_i \right\|_{\mathbb{X}}^2 = \min_{\mathbb{W} \in \mathcal{V}_N^{\mathbb{X}, \mathcal{N}}} \sum_{i=1}^{n_s} \left\| \mathbf{u}_i - \mathbb{W} \mathbb{W}^T \mathbb{X} \mathbf{u}_i \right\|_{\mathbb{X}}^2 = \sum_{i=N+1}^r \sigma_i^2$$

Computational point of view

Computationally

$$\tilde{\mathbb{Z}}^T \tilde{\mathbb{Z}} \tilde{\psi}_i = \sigma_i^2 \tilde{\psi}_i, \quad i = 1, \dots, r$$

if $n_s < \mathcal{N}$ we can conveniently obtain the POD basis without forming the matrix $\mathbb{X}^{1/2}$.

Indeed,

- first compute the correlation matrix $\tilde{\mathbb{C}} = \tilde{\mathbb{Z}}^T \tilde{\mathbb{Z}} = \mathbb{Z}^T \mathbb{X} \mathbb{Z}$ and its first N eigenvectors $\tilde{\psi}_1, \dots, \tilde{\psi}_N$
- define the POD basis as $\mathbb{V} = [\zeta_1 | \dots | \zeta_N]$, where

$$\zeta_i = \mathbb{X}^{-1/2} \tilde{\zeta}_i = \mathbb{X}^{-1/2} \frac{1}{\sigma_i} \tilde{\mathbb{Z}} \tilde{\psi}_i = \frac{1}{\sigma_i} \mathbb{Z} \tilde{\psi}_i.$$

Algorithm

Proper Orthogonal Decomposition (POD) Algorithm for the energy inner product

- 1: Compute $\mathbb{V} = \text{POD}(\mathbb{Z}, \mathbb{X}, \varepsilon_{\text{POD}})$
- 2: form the correlation matrix $\tilde{\mathbb{C}} = \mathbb{Z}^T \mathbb{X} \mathbb{Z}$
- 3: solve the eigenvalue problem $\tilde{\mathbb{C}} \tilde{\psi}_i = \sigma_i^2 \tilde{\psi}_i, \quad i = 1, \dots, r$
- 4: set $\zeta_i = \frac{1}{\sigma_i} \mathbb{Z} \tilde{\psi}_i$
- 5: define N as the minimum integer such that $l(N) \geq 1 - \varepsilon_{\text{POD}}$
- 6: set $\mathbb{V} = [\zeta_1 | \dots | \zeta_N]$

POD basis and approximation property

The POD basis is, among all basis of dimension $N < n_s$, the one which best approximates the set of solution snapshots

$$\mathcal{M}_{n_s} = \{u_h(\boldsymbol{\mu}^1), \dots, u_h(\boldsymbol{\mu}^{n_s})\}.$$

However there remains two questions:

- 1 the approximation property of the POD basis with respect to the entire solutions set $\mathcal{M} = \{u_h(\boldsymbol{\mu}) : \boldsymbol{\mu} \in \mathcal{D}\}$
- 2 how to select the parameter samples $\Xi_S = \{\boldsymbol{\mu}^1, \dots, \boldsymbol{\mu}^{n_s}\}$ so that the corresponding snapshots set is sufficiently representative of the solutions set.

A continuous formulation

To this end, it is useful to introduce a $\mathcal{D}$ -continuous version of the POD technique.

Indeed, if we are interested to find a POD basis of dimension N that approximates the entire solutions set $\mathcal{M}$, we have to consider the following minimization problem

$$\min_{\mathbb{W} \in V_N} \int_{\mathcal{D}} \|u_h(\mu) - \mathbb{W}^T \mathbb{W}^T u_h(\mu)\|_2^2 d\mu$$

A continuous formulation: Operators I

Let us suppose that $u_h(\mu) \in L^2(\mathcal{D}; \mathbb{R}^{\mathcal{N}})$, i.e.

$$\int_{\mathcal{D}} \|u_h(\mu)\|_2^2 d\mu < \infty;$$

$u_h(\mu)$ is called a Hilbert-Schmidt kernel. A continuous analogue of the snapshots matrix $\mathbb{Z}$ is given by the operator $T : L^2(\mathcal{D}) \rightarrow \mathbb{R}^{\mathcal{N}}$ such that

$$Tg = \int_{\mathcal{D}} u_h(\mu)g(\mu)d\mu \quad \forall g \in L^2(\mathcal{D}).$$

Its adjoint $T^* : \mathbb{R}^{\mathcal{N}} \rightarrow L^2(\mathcal{D})$ is defined as

$$(g, T^*w)_{L^2(\mathcal{D})} = (Tg, w)_2 \quad \forall g \in L^2(\mathcal{D}), w \in \mathbb{R}^{\mathcal{N}}.$$

A continuous formulation: Operators II

As a result

$$T^*w = (u_h(\boldsymbol{\mu}), w)_2 \quad \forall w \in \mathbb{R}^{\mathcal{N}},$$

that is, T^* is the continuous analogue of the matrix $\mathbb{Z}^T$. Note that $Tg \in \mathbb{R}^{\mathcal{N}}$ and $\text{rank}(T) = r \leq \mathcal{N}$, while $T^*w = (T^*w)(\boldsymbol{\mu}) \in L^2(\mathcal{D})$.

Continuous Formulation I

Introducing the Hilbert Schmidt norm on an operator $T : L^2(\mathcal{D}) \rightarrow \mathbb{R}^{\mathcal{N}}$

$$\|T\|_{HS}^2 = \|u_h(\mu)\|_{L^2(\mathcal{D}; \mathbb{R}^{\mathcal{N}})}^2$$

Since T is compact, $K = TT^* : \mathbb{R}^{\mathcal{N}} \rightarrow \mathbb{R}^{\mathcal{N}}$ and

$C = T^*T : L^2(\mathcal{D}) \rightarrow L^2(\mathcal{D})$ are self-adjoint, non-negative, compact operators, given by

$$\begin{aligned} Cg &= \int_{\mathcal{D}} (u_h(\mu), u_h(\mu'))_2 g(\mu') d\mu' \quad \forall g \in L^2(\mathcal{D}), \\ Kw &= \int_{\mathcal{D}} u_h(\mu) (u_h(\mu), w)_2 d\mu \quad \forall w \in \mathbb{R}^{\mathcal{N}}. \end{aligned}$$

Note that the operators K and C are nothing but the $\mathcal{D}$ -continuous analogue of the correlation matrices $\mathbb{K} = \mathbb{Z}\mathbb{Z}^T$ and $\mathbb{C} = \mathbb{Z}^T\mathbb{Z}$.

Continuous Formulation II

Actually, since K is a linear map from $\mathbb{R}^{\mathcal{N}}$ to $\mathbb{R}^{\mathcal{N}}$, it is represented by the $\mathcal{N} \times \mathcal{N}$ symmetric positive definite matrix

$$K = \int_{\mathcal{D}} u_h(\mu) u_h^T(\mu) d\mu$$

whose eigenvalues $\sigma_1^2 \geq \dots \geq \sigma_r^2 \geq 0$ and associated orthonormal eigenvectors $\zeta_i \in \mathbb{R}^{\mathcal{N}}$ satisfy

$$K\zeta_i = \sigma_i^2 \zeta_i, \quad i = 1, \dots, r.$$

Moreover, the functions $\psi_i \in L^2(\mathcal{D})$ defined by

$$\psi_i = \frac{1}{\sigma_i} T^* \zeta_i, \quad i = 1, \dots, r,$$

are the eigenvectors of C . Finally, as $u_h(\mu)$ admits the expansion

$$u_h(\mu) = \sum_{i=1}^r \sigma_i \zeta_i \psi_i(\mu) = \sum_{i=1}^r \zeta_i (u_h(\mu), \zeta_i)_2$$

Continuous Formulation III

the following decomposition holds for T

$$T = \sum_{i=1}^r \sigma_i \zeta_i(\psi_i(\boldsymbol{\mu}), \cdot)_{L^2(\mathcal{D})} \quad (5)$$

As in the matrix context, it can be proved that

$$\|T\|_{\mathcal{L}(L^2(\mathcal{D}), \mathbb{R}^{\mathcal{N}})} = \sigma_1, \quad \|T\|_{HS} = \sqrt{\sum_{i=1}^r \sigma_i^2}.$$

Moreover, truncating the sum 5 to the first N terms, we obtain the best rank N approximation to the operator T .

Continuous formulation: Schmidt theorem

We have the following theorem

Theorem (Schmidt⁴)

The operator $T_N : L^2(\mathcal{D}) \rightarrow \mathbb{R}^{\mathcal{N}}$ defined by

$$T_N = \sum_{i=1}^N \sigma_i \zeta_i (\psi_i(\boldsymbol{\mu}), \cdot)_{L^2(\mathcal{D})}, \quad 0 \leq N < r$$

satisfies the following optimality property

$$\|T - T_N\|_{HS} = \min_{B \in \mathcal{B}_N} \|T - B\|_{HS} = \sqrt{\sum_{i=N+1}^r \sigma_i^2},$$

where $\mathcal{B}_N = \{B \in \mathcal{L}(L^2(\mathcal{D}), \mathbb{R}^{\mathcal{N}}) : \text{rank}(B) \leq N\}$.

⁴see proof in Stewart, G.W.: On the early history of the singular value decomposition. SIAM review 35(4), 551–566 (1993)

The previous theorem provides a solution to the minimization problem , as stated in the following Proposition.

Proposition

The POD basis $\mathbb{V} = [\zeta_1 | \dots | \zeta_N] \in \mathbb{R}^{\mathcal{N} \times N}$ is such that

$$\int_{\mathcal{D}} \left\| u_h(\mu) - \mathbb{V} \mathbb{V}^T u_h(\mu) \right\|_2^2 d\mu = \min_{\mathbb{W} \in \mathcal{V}_N} \int_{\mathcal{D}} \left\| u_h(\mu) - \mathbb{W}^T \mathbb{W}^T u_h(\mu) \right\|_2^2 d\mu$$

. Moreover

$$\int_{\mathcal{D}} \left\| u_h(\mu) - \mathbb{V} \mathbb{V}^T u_h(\mu) \right\|_2^2 d\mu = \sum_{i=N+1}^r \sigma_i^2$$

Proof Since $\psi_i = \frac{1}{\sigma_i} T^* \zeta_i$, thanks to the definition of T^* we can express T_N as

$$\begin{aligned} T_N g &= \sum_{i=1}^N \sigma_i \zeta_i (\psi_i(\mu), g(\mu))_{L^2(\mathcal{D})} = \int_{\mathcal{D}} (u_h(\mu), \zeta_i)_2 \zeta_i g(\mu) d\mu \\ &= \int_{\mathcal{D}} \mathbb{V} \mathbb{V}^T u_h(\mu) g(\mu) d\mu \quad \forall g \in L^2(\mathcal{D}). \end{aligned}$$

Then

$$\|T - T_N\|_{HS}^2 = \|T - \mathbb{V}\mathbb{V}^T T\|_{HS}^2 = \int_{\mathcal{D}} \|u_h(\mu) - \mathbb{V}\mathbb{V}^T u_h(\mu)\|_2^2 d\mu$$

Since

$$\|T - T_N\|_{HS}^2 = \min_{B \in \mathcal{B}_N} \|T - B\|_{HS}^2 \leq \min_{\mathbb{W} \in \mathcal{V}_N} \|T - \mathbb{W}\mathbb{W}^T T\|_{HS}^2,$$

and $T_N = \mathbb{V}\mathbb{V}^T T$, we conclude that

$$\|T - \mathbb{V}\mathbb{V}^T T\|_{HS}^2 = \min_{\mathbb{W} \in \mathcal{V}_N} \|T - \mathbb{W}\mathbb{W}^T T\|_{HS}^2$$

□

Back to discrete problem I

Thanks to the previous analysis of the $\mathcal{D}$ -continuous version of the POD we can deal with the questions concerning approximation and sampling properties of POD basis functions. The key is to regard the discrete minimization problem as an approximation of the continuous minimization problem. To this end, we introduce a suitable quadrature formula

$$\int_{\mathcal{D}} f(\boldsymbol{\mu}) d\boldsymbol{\mu} \approx \sum_{i=1}^{n_s} w_i f(\boldsymbol{\mu}^i)$$

to approximate the integrals over $\mathcal{D}$, for any continuous function $f : \mathcal{D} \rightarrow \mathbb{R}$, where $w_i > 0$ and $\boldsymbol{\mu}^i$ are suitable quadrature weights and quadrature points, respectively.

Then,

$$\int_{\mathcal{D}} \left\| \mathbf{u}_h(\boldsymbol{\mu}) - \mathbb{W}\mathbb{W}^T \mathbf{u}_h(\boldsymbol{\mu}) \right\|_2^2 d\boldsymbol{\mu} \approx \sum_{i=1}^{n_s} w_i \left\| \mathbf{u}_h(\boldsymbol{\mu}^i) - \mathbb{W}\mathbb{W}^T \mathbf{u}_h(\boldsymbol{\mu}^i) \right\|_2^2.$$

Back to discrete problem II

The parameter samples location are thus defined by a suitable quadrature formula in the parameter space. The addition to the POD algorithm is the weighting coefficients w_i .

However, by defining $\mathbb{D} = \text{diag}(w_1, \dots, w_{n_s}) \in \mathbb{R}^{n_s \times n_s}$,

$$\sum_{i=1}^{n_s} w_i \left\| \mathbf{u}_h(\boldsymbol{\mu}^i) - \mathbb{W} \mathbb{W}^T \mathbf{u}_h(\boldsymbol{\mu}^i) \right\|_2^2 = \left\| \mathbb{Z} \mathbb{D}^{1/2} - \mathbb{W} \mathbb{W}^T \mathbb{Z} \mathbb{D}^{1/2} \right\|_F^2.$$

The POD basis associated to the snapshots set

$\mathcal{M}_{n_s} = \{ \mathbf{u}_h(\boldsymbol{\mu}^1), \dots, \mathbf{u}_h(\boldsymbol{\mu}^{n_s}) \}$ is given by $\mathbb{V} = [\boldsymbol{\zeta}_1 | \dots | \boldsymbol{\zeta}_N]$, with $\boldsymbol{\zeta}_i = \frac{1}{\sigma_i} \mathbb{Z} \mathbb{D}^{1/2} \tilde{\boldsymbol{\psi}}_i$ and

$$\tilde{\mathbb{Z}}^T \tilde{\mathbb{Z}}_{\boldsymbol{\psi}}^i = \sigma_i^s \tilde{\boldsymbol{\psi}}_i, \quad \text{with} \quad \tilde{\mathbb{Z}} = \mathbb{Z} \mathbb{D}^{1/2}.$$

Algorithm: POD with energy norm and quadrature weights

POD Algorithm for the energy inner product and quadrature weights

- 1: Compute $\mathbb{V} = \text{POD}(\mathbb{Z}, \mathbb{X}, \mathbb{D}, \varepsilon_{\text{POD}})$
- 2: Set $\tilde{\mathbb{Z}} = \mathbb{Z}\mathbb{D}^{1/2}$
- 3: form the correlation matrix $\tilde{\mathbb{C}} = \tilde{\mathbb{Z}}^T \mathbb{X} \tilde{\mathbb{Z}}$
- 4: solve the eigenvalue problem $\tilde{\mathbb{C}} \tilde{\psi}_i = \sigma_i^2 \tilde{\psi}_i, \quad i = 1, \dots, r$
- 5: set $\zeta_i = \frac{1}{\sigma_i} \mathbb{Z} \tilde{\psi}_i$
- 6: define N as the minimum integer such that $l(N) \geq 1 - \varepsilon_{\text{POD}}$
- 7: set $\mathbb{V} = [\zeta_1 | \dots | \zeta_N]$

In practice, if you choose eg a random sampling you have to scale the correlation matrix by $1/n_s$

Exercises

- ① Show that $\mathbb{A}^T \mathbb{A} \mathbf{v}_i = \sigma_i^2 \mathbf{v}_i$, $\mathbb{A}^T \mathbf{u}_i = \sigma_i^2 \mathbf{u}_i$ for any $i = 1, \dots, n$, whence $\lambda_i(\mathbb{A}^T \mathbb{A}) = \lambda_i(\Sigma^2) = (\sigma_i(\mathbb{A}))^2$.
- ② Show that if $\mathbb{A} \in \mathbb{R}^{n \times n}$ is symmetric with eigenvalues given by $\lambda_1(\mathbb{A}) \geq \lambda_2(\mathbb{A}) \geq \dots \geq \lambda_n(\mathbb{A})$, then $\sigma_i(\mathbb{A}) = |\lambda_i(\mathbb{A})|$.
- ③ Show that if $\mathbb{A} \in \mathbb{R}^{m \times n}$, then

$$\sigma_{\max}(\mathbb{A}) = \max_{\mathbf{y} \in \mathbb{R}^m, \mathbf{x} \in \mathbb{R}^n} \frac{\mathbf{y}^T \mathbb{A} \mathbf{x}}{\|\mathbf{x}\| \|\mathbf{y}\|}.$$